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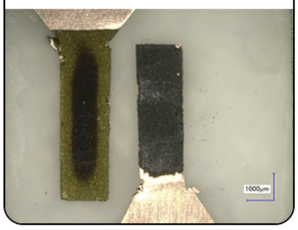

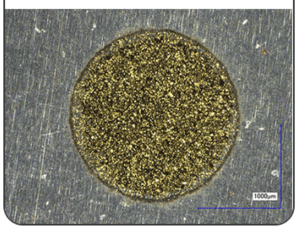

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Analysis of the Implications of Rapid Charging on Lithium-Ion Battery Performance

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Fast charging is critical for efficient electrical vehicle operation. In this study, the fast charging performance limitations are elucidated with emphasis on the influence of temperature extremes. The role of charging protocols and electrode design parameters on charging time reduction are analyzed. Conventional charging protocol at high C-rate (e.g. at 3C) demonstrates that the performance limitation in rapid charging primarily originates from lithium ion transport in the electrolyte (electrolyte resistance) at moderate and high operating temperatures. However, interfacial kinetics resistance becomes the limiting mechanism for the 1C charging rate at low temperatures. To overcome the low temperature limitation, self-heating has been found to boost the cell performance and coulombic efficiency. An effective charging protocol has been suggested by maximizing lithium concentration at the anode/separator interface. Based on the advantageous combination of charging protocol and temperature, we propose a pulse-charging protocol with adiabatic operation, which can reduce charging time and increase performance. Furthermore, appropriate electrode design, such as reducing the electrode thickness and increasing the porosity, results in improved charging performance.

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The detrimental impact of greenhouse gas emissions along with the increase of petroleum prices have urged the researching for cleaner and cheaper alternatives. The hybrid electric vehicles (HEV) and the pure electric vehicles (EV) attracted significant attention as an alternative transportation source. The high power and energy density of the lithium ion batteries (LIBs) have made it a suitable candidate to HEV and EV. However, these batteries have practical barriers, which impede their prevalence to the current market. One of these barriers is the prolonged charging time of these batteries.[1–3]

Fast charging is a terminology used for a technique that provide a charge duration of less than 1 hour with a charge rate higher than 1C.[4] The trade-off between the charging time and efficiency is an imperative concern for fast charging. The charging time of the LIBs is affected by (1) the lithiation/delithiation rate (2) the ionic transport in electrolyte (3) the ion diffusion in the active materials. From the ion transport mechanism, the lithium ion undergoes several electrochemical resistances, which are the charge transfer kinetics resistance, electrolyte resistance, solid electrolyte interphase (SEI) film resistance, and solid-state diffusion resistance. These resistances hinder the Li^+ to travel smoothly, which extends the charging time of the cell. According to previous studies, the charge performance can be improved by charging protocol, operation condition and electrode microstructure.

Several charging protocols have been proposed to achieve shorter charging times. Notten et al.[5] proposed a boostcharging protocol where a boostcharge period is applied to the battery followed by the conventional Constant-Current Constant-Voltage (CCCV) charging. The boostcharging protocol fulfilled the fast charging requirements with little degradation effects. In a different work, Purushothaman and Landau[6] developed a macro-homogeneous lithium diffusion model of lithium into the graphite anode for pulse charging protocol. The model predicts that the charging time with pulse charging could achieve about 2.5 faster than conventional charging. Moreover, Chung et al.[7] developed a linearly descending current charging protocol based on a simple continuous non-porous electrode model. It showed that the charging time is reduced when charging with a linearly descending current protocol compared to a constant current charging protocol. However, Sikha et al.[8] showed that the decrease of charging time accompanied with the overcharging of cell that harmfully affects the cycle life of the battery. To solve the overcharging problem, Sikha et al.[8] proposed a charging protocol that vary the current during the linearly descending current charging protocol. Later, Anseán et al.[9] proposed a multistage fast charging technique for high power LFP cells. The cycle-life and performance of the cells were further improved.

Controlling the operation temperature is also a way to improve the charge performance. Zhang et al.[10,11] studied the charge and discharge characteristics of lithium ion cells using various electrochemical techniques at different temperatures. They concluded that, the electrolyte resistance dominates the cell resistance at room temperatures ($\sim$25°C). Under low temperature operation condition ($<$0°C), the interfacial kinetics resistance, which is related to the intercalation process, dominates the cell resistance because temperature has larger influence on the electrochemical reaction rate than the ionic conductivity and diffusivity. Fan and Tan[12] also studied the charging process of lithium ion cells at low temperatures with both conventional and pulse charging. Differently, they concluded that lithium solid diffusion inside the graphite is the rate-limiting factor in charging at low temperatures. The difference of the conclusions is possible due to the self-heating of cell. The increase of temperature changes the dominated cell resistance. The heat generation is a result of the high current densities used for fast charging at moderate and high temperatures.[13–15] For subzero temperatures however, the high cell resistance results in the high heat generation.[16] The charging process of cell under low temperature operation with the self-heating will be further discussed in this study. Beside the charging protocol and temperature effect, the electrode microstructure is suggested to be a crucial factor for charging rate according to many studies.[17] Park et al.[18] investigated the effect of different design parameters (i.e. porosity and thickness of electrode) on the charging time of the cell. Among the studied parameters, it was found that thinner cathode thickness would considerably reduce the charging time of the cell.

Investigations of fast charging issue are limited despite its significance intended for EV applications. The lack of understanding of the basic mechanism and performance of the LIB during fast charging is one of the overriding barriers, since the investigations and studies have been mostly limited by experimental measurements and findings. Few modeling studies have attempted to investigate the fast charging operation of LIBs, despite the 2D pseudo electrochemical model of Li-ion cells[19] has achieved a noteworthy success in the prediction of the LIBs performance and behavior compared with experimental results. In our study, due to the fact that the electrochemical processes inside the battery are temperature dependent, a fully electrochemical thermal model has been implemented.[20–23] By using the electrochemical/thermal coupled model the basic mechanism of the fast charging of Li-ion cells at subzero, moderate and high temperatures are investigated. Besides the temperature effect, the influence of different charging protocols and physical electrode properties and design considerations thereof are investigated.

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Model and Theory

In this study, a typical 18650 cylindrical Li-ion cell with a nominal capacity of 2.2 Ah is used as a base case for the mathematical model.[24] The cell consists of a graphite anode on a copper current collector, and a $LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$ (NCM) cathode, on a carbon-coated aluminum current collector.

The cell is filled with 1.2 M $LiPF_6$ in Propylene Carbonate-Ethylene Carbonate-Dimethyl Carbonate EC-EMC-DMC (27-10-63 vol. %) electrolyte. In the present work, AutoLion-1D software is used, which is based on the 1D electrochemical/thermal (ECT) coupled model outlined in Kalupson et al.[25] The pertinent governing equations are presented below. The charge conservation equation can be expressed as in Eq. 1, where the specific surface area and the effective electrical conductivity of the solid phase are given in Eq. 2:

$$\nabla.\left(\sigma_{s,i}^{eff}\nabla\phi_s(\mathrm{x,t})\right) = -a_{s,i}\, j_{loc,i} \quad [1]$$

$$a_{s,i} = \frac{3\varepsilon_{s,i}}{R_{Particle,i}};\quad \sigma_{s,i}^{eff} = \sigma_{s,i}\varepsilon_{s,i}^{\gamma} \quad [2]$$

The charge conservation in the electrolyte can be expressed in Eq. 3.

$$\nabla.\left(\kappa_e^{eff}\nabla\phi_e(x,t) + \kappa_D^{eff}\nabla\ln c_e(\mathrm{x,t})\right) = -a_{s,i}\, j_{loc} \quad [3]$$

where the effective ionic conductivity κ_e^{eff} and the effective diffusional ionic conductivity κ_D^{eff} is given in Eq. 4 and 5 respectively.[26]

$$\kappa_e^{eff} = 10^{-4}\times c_e\begin{pmatrix} -10.5 + 6.68\times10^{-4}c_e +\\ 4.94\times10^{-5}c_e + 0.047T -\\ 1.78\times10^{-5}c_eT - 8.86\times10^{-10}c_e^2T\\ -6.96\times10^{-5}T^2 + 2.8\times10^{-8}c_eT^2\end{pmatrix}^2 \varepsilon_i^{\gamma} \quad [4]$$

$$\kappa_D^{eff} = \frac{2RT\kappa_e^{eff}}{F}\left(1 + \frac{d\ln f}{d\ln c_e}\right)\left(t^+ - 1\right) \quad [5]$$

The mass conservation of Li^+ in the active material is described by Fick's law as expressed in Eq. 6. The boundary condition of mass conservation equation is given in Eq. 7.

$$\frac{\partial c_s(\mathrm{x},r,t)}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(D_{s,i}r^2\frac{\partial c_s(\mathrm{x},r,t)}{\partial r}\right) \quad [6]$$

$$-D_{s,i}\left.\frac{\partial c_{s,i}(\mathrm{x},r,t)}{\partial r}\right|_{r=R_i} = \frac{j_{loc,i}}{F} \quad [7]$$

The mass conservation of Li^+ in the electrolyte is expressed in Eq. 8.

$$\varepsilon_{e,i}\frac{\partial c_e(x,t)}{\partial t} = \nabla.\left(D_e^{eff}\nabla c_e(x,t)\right) + \frac{a_{s,i}\, j_{loc,i}}{F}\left(1 - t^+\right) \quad [8]$$

The lithiation and delithiation reaction rates (charge transfer kinetics) follow the Butler-Volmer equation given in Eq. 9.

$$j_{loc,i} = j_{0,i}\left[\exp\left(\frac{\alpha_{a,i}\eta_i F}{RT}\right) - \exp\left(\frac{-\alpha_{c,i}\eta_i F}{RT}\right)\right] \quad [9]$$

with η the kinetic overpotential which is given by:

$$\eta_i = \phi_{s,i}(x,t) - \phi_{e,i}(x,t) - U_{ref,i} - j_{loc,i}R_f \quad [10]$$

In Eq. 9, the exchange current density $j_{0,i}$ is given as,

$$j_{0,i} = k_i(T)c_e^{\alpha_a}(\mathrm{x,t})\left(c_{s,\max} - c_{s,i}(x,R,t)\right)^{\alpha_a}c_{s,i}^{\alpha_c}(x,R,t) \quad [11]$$

The open circuit potential of the graphite anode and the NCM cathode are obtained from experimental measurements from previous studies and the resulting empirical formulas are given in Eq. 12 and 13 simultaneously.[27,28]

$$U_n[\chi] = \begin{pmatrix} 0.1493 + 0.8493e^{-61.79\chi} +\\ 0.3824e^{-665.8\chi} - e^{-39.42\chi - 41.92} -\\ 0.03131\arctan(25.59\chi - 4.099) -\\ 0.009434\arctan(32.49\chi - 15.74)\end{pmatrix}(0 \le \chi \le 1) \quad [12]$$

$$U_p[\chi] = \begin{pmatrix} -10.72\chi^4 + 23.88\chi^3 -\\ 16.77\chi^2 + 2.595\chi + 4.563\end{pmatrix}(0.3 \le \chi \le 1) \quad [13]$$

where χ is the state of charge.

The energy conservation in the cell is based on the thermal lumped model and is formulated as given in Eq. 14.

$$m_{cell}C_p\frac{\partial T}{\partial t} = \left(\dot{Q}_{ohm} + \dot{Q}_{rxn} + \dot{Q}_{act}\right) - h_{conv}A_{cell}\left(T - T_{amb}\right) \quad [14]$$

In the individual heat generation, the first term on the right hand side of Eq. 14 can be expressed as Eq. 15, and the second term presents the convection heat transfer to the ambient.

$$\begin{aligned} \dot{Q}_{ohm} &= A_e\int_0^L\Big(\sigma_{s,i}^{eff}\nabla\phi_s(x,t)\cdot\nabla\phi_s(x,t) + \kappa_e^{eff}\nabla\phi_e(x,t)\cdot\nabla\phi_e(x,t)\\ &\qquad + \kappa_D^{eff}\nabla\ln c_e(x,t).\nabla\phi_e(x,t)\Big)dx\\ \dot{Q}_{rxn} &= A_e\int_0^L a_{s,i}\, j_{0,i}\left(T\frac{dU_i}{dT}\right)dx\\ \dot{Q}_{act} &= A_e\int_0^L a_{s,i}\, j_{0,i}\left(\phi_s(x,t) - \phi_e(x,t) - U_i\right)dx \end{aligned} \quad [15]$$

In the model, an Arrhenius dependence is used to update the temperature dependent parameters as given by Eq. 16, where P represents any temperature dependent parameter used in the present model, which includes $\sigma_{s,i}^{eff}$, $D_{s,i}$, k_i, and $U_{ref,i}$. One should note that the ionic conductivity κ_e^{eff} is directly calculated from Eq. 4.

$$P_i = P_{0,i}\exp\left(\frac{E_{a,i}}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right) \quad [16]$$

The baseline design parameters and electrochemical parameters used for the present model are given in Table I. Figure 1 shows a comparison between the modeling results predicted by the present model and experimental results under different temperature and C-rate.[16] The simulation results show good agreement with the experimental results under high C-rate and low temperature. In the present study, the experimentally validated model is used to explore the operation and performance of LIB fast charging. It is important to note that as a first approximation, the influence of electrochemical double layer (EDL) on the charging behavior[29,30] and the thermal induced degradation are not taken into account. The effects of the existence of the EDL and thermal degradation are left as a future exercise.

Results and Discussion

Fast charging performance.— During charging, the accompanying charge transport resistances include the ionic resistance, electrical resistance, and interfacial kinetics resistance. Since the limitation of fast charge is mainly due to the resistance inside the cell, which hinders the ion transport, study the resistance inside the cell can help us to improve the charge performance. In fast charging process, our objective is to decrease the charging time, but maintain the coulombic efficiency.

In this section, the cell performances during fast charging are investigated under different temperature and charging current. The CCCV charging protocol is employed in the simulations. In this work, 100%

Table I. Design and electrochemical parameters of the base simulation.

Design Parameters	Anode	Separator	Cathode
Thickness (μm)	81	20	78
Porosity	0.26	0.46	0.28
Loading (mAh/cm^2)	4.5		3.9
Electrolyte concentration (mol dm^{-3})	1.2		
Particle radius (μm)	10		5
Exchange current density i_0 (A m^{-2})	12[23] ($x = 0.5$)		2[16] (y = 0.5)
Activation energy of i_0 (kJ mol^{-1})	68[37]		50[37]
Charge transfer coefficient α_a, α_c	0.5, 0.5		0.5, 0.5
Film resistance R_f (Ω cm^{-2})	10[16]		10[16]
Activation energy of R_f (kJ mol^{-1})	50[16]		50[16]
Solid state diffusivity D_s (m^2/s)	1.6×10^{-14}[20]		1.0×10^{-10}[20]
Activation energy of D_s (kJ mol^{-1})	30[38]		30[16]
Contact resistance (Ω cm^2)	6[16]		
Heat Transfer coefficient (W/m^2 K)	28.4[16]		

charging efficiency refers to the capacity when the battery was fully charged with constant-current constant voltage protocol (CCCV) at 25°C and 1C during the constant-current process. During the CCCV charging protocol, the cell was first charged with constant current until the voltage reached 4.2 V. After the cell voltage reached 4.2 V, the cell was charged under constant voltage (4.2 V) until the current is less than 0.05 A. The operation temperatures of −20°C, 25°C, and 45°C, are chosen to represent the subzero, moderate, and high temperature operation condition respectively. To include the self-heating, a comparison between the isothermal, adiabatic, and convective cooling condition is outlined to analyze the thermal influence on the cell resistance. The conditions for the base case for fast charging at moderate and high operating temperatures are chosen to be 3C-rate.[18,31] However, for subzero temperatures, we chose 1C-rate as the base case for fast charging since the CC period is very short (Zhao et al.[32]) compared to the charging process at moderate and high operating temperatures.

Figure 2 shows the charge time and coulombic efficiency over a wide range of charging/discharging rates. The charging time corresponds to the time that the cell takes to finish the CCCV charging protocol. In the case of moderate (25°C) and high temperatures (45°C), the cell is charged with range of 1C to 4C followed by galvanostatic discharge at the same rates. For the subzero temperature (−20°C), the cell is charged with range of 0.2C to 1C rates and discharged galvanostatically at the same rate. Theoretically, high charging/discharging C-rates should be avoided at low temperatures, because the low conductivity and diffusivity (due to the low temperature) enhance the potential and concentration polarization, which brings the cell to (constant-voltage) CV stage rapidly. The decrease of the duration of (constant-current) CC period increase the charging time compare to moderate and high temperature as shown in Figure 2. For $T = 25$°C and 45°C, the increase of charging rates can reduce the charging time. One should notice that, when the charging rate in the CC period beyond 3C, the increasing charging rate does not reduce the charging time significantly for the adiabatic and convention cooling conditions. At high charging rates (C-rate > 3C), the CV period is extended, which does not decrease the charging time significantly according to the study of Zhao et al.[32] In addition, although we can decrease the charging time by increasing the charging rate, the reduction in the charging time is accompanied with the expense of the columbic efficiency.

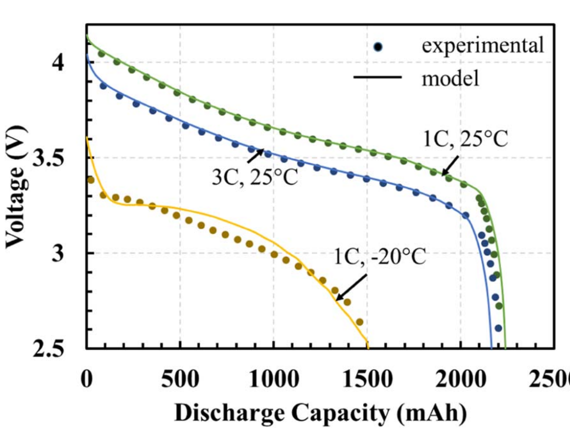

Figure 1. Comparison between modeling and experimental data.[16]

The results (Figure 2) also demonstrate the sensitivity and discrepancy in the performance of the cell at different thermal conditions (adiabatic, convention cooling, and isothermal). A higher deterioration in the cell performance is noticed in the isothermal conditions in terms of the charging time and efficiency compared to other thermal conditions. From Figure 2, it is clear that the adiabatic conditions at all operating temperatures outperform the convention cooling and isothermal conditions. The cell performance enhancement is mainly due to the higher temperature increase under the adiabatic condition as compared to the other thermal conditions as shown in Figure 3. To better understand the cell behavior with different thermal condition, the examination of cell resistance is conducted. The cell resistance at different thermal conditions are identified and compared with the base case thermal condition. The cell resistance are calculated according to the methodology developed by Ji et al.[16] Figure 4 presents the cell resistance as a function of charging time for three different operating temperatures with the CCCV charging protocol. First, for charging at 25°C (Figure 4(a)), the electrolyte resistance is the dominated resistance and increases during the charging process. The lower porosity and larger thickness of anode makes the electrolyte resistance is higher than the cathode side. Figure 5 shows the electrolyte concentration and potential at the end of the CC charging period. The electrolyte concentration at the cathode current collector reaches about 1.5 times higher than the initial concentration at the end of the CC period. However, the large electrolyte potential drop decreases the duration of CC period. For the CV period, the electrolyte resistance keeps increasing with charging time as shown in Figure 4(a) due to the decrease of applied current. The charging current decreases through the charging process is because of the increase of lithium concentration in the active material and the decrease of lithium concentration in the electrolyte. Figure 4(b) shows the cell resistance for the CCCV charging at 45°C. The predicted charging performance at 45 °C shows generally analogous trend to 25°C as shown in Figure 2(c) and 2(d). Because of the high temperature, the resistance at 45°C is lower than the one at 25°C. The range of predicted charging time and coulombic efficiency in the cell are also better than those observed at 25°C.

Figure 4(c) demonstrates the cell resistance during slow charging at 0.5C when $T = 25$°C. In comparison between fast and slow charging, the anode particle resistance has two peaks, which related to the

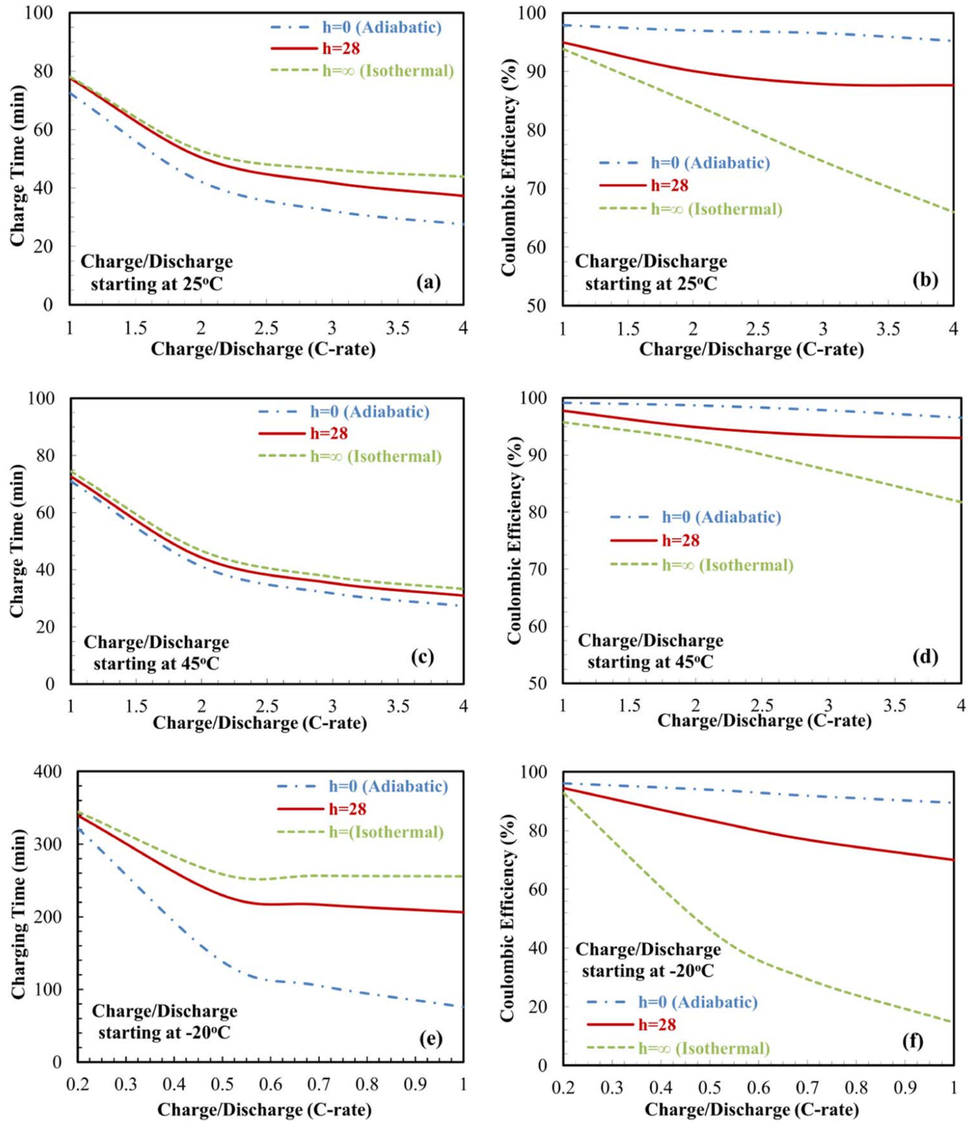

Figure 2. The charging time and the columbic efficiency as a function of charge\discharge C-rates with different thermal conditions (heat transfer coefficients), adiabatic condition ($h = 0$ W/m^2.K), natural convection (h = 28 W/m^2.K) and isothermal condition ($h \rightarrow \infty$ W/m^2.K). (a), (b) the charging time and columbic efficiency respectively with an ambient temperature of 25°C the charge\discharge range is 1C-4C. (c), (d) the ambient temperature is 45°C and the charge\discharge range is 1C-4C. (e), (f) the ambient temperature is −20°C and the charge\discharge range is 0.2C-1C.

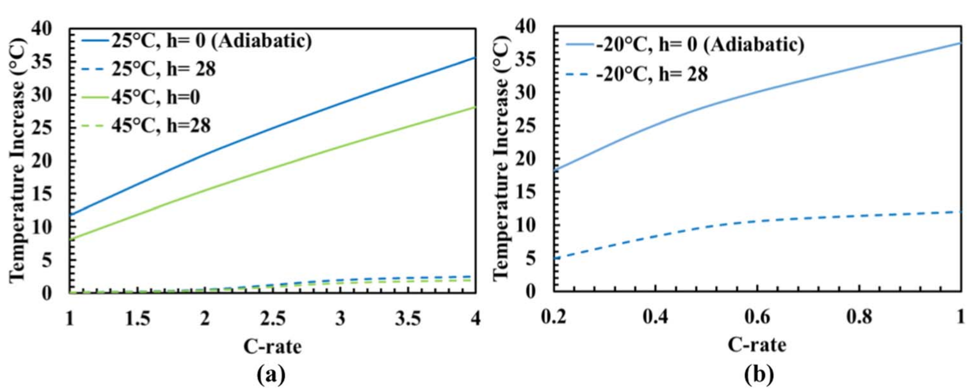

Figure 3. The temperature increase after the CC charging process under different thermal conditions ($h = 0$ W/m^2.K) and natural convection ($h = 28$ W/m^2.K) (a) The ambient temperature is 25°C and 45°C, and the charge range is 1C-4C. (b) The ambient temperature is −20°C and the charge range is 0.2C-1C.

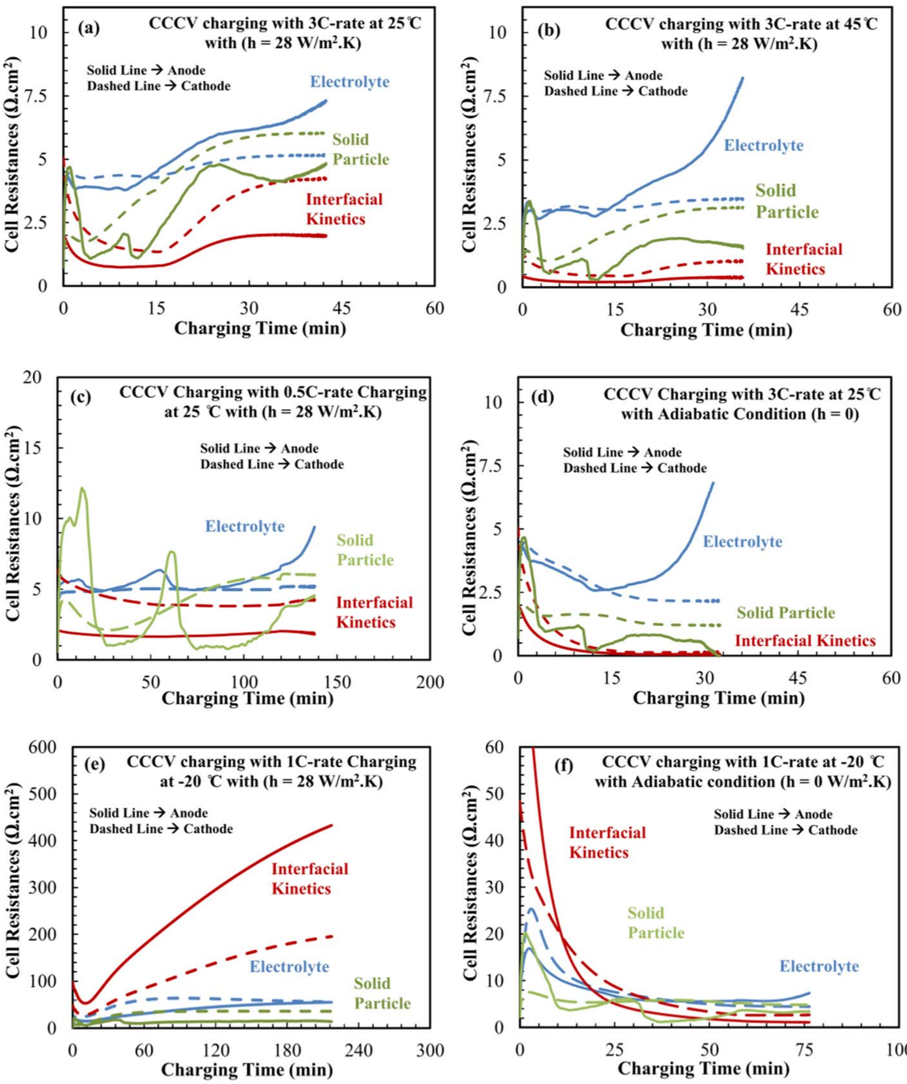

Figure 4. The evolution of the battery internal resistances during conventional (CCCV) charging in the anode and the cathode as a function charging time at different operating temperature. (a) Fast charging starting at 25°C with charging rate of 3C and a heat transfer coefficient of 28 $W/m^2 \cdot K$. (b) Fast charging starting at 45°C with charging rate of 3C and a heat transfer coefficient of 28 $W/m^2 \cdot K$. (c) Slow charging starting at 25°C with charging rate of 0.5C and a heat transfer coefficient of 28 $W/m^2 \cdot K$. (d) Fast charging starting at 25°C with charging rate of 3C and a heat transfer coefficient of 0 $W/m^2 \cdot K$. (e) Fast charging starting at -20°C with charging rate of 1C and a heat transfer coefficient of 28 $W/m^2 \cdot K$. (f) Fast charging starting at -20°C with charging rate of 1C and a heat transfer coefficient of 0 $W/m^2 \cdot K$.

OCP of graphite that infers a uniform utilization of graphite particles at slow charging. In addition, the electrolyte resistance shows a more uniform resistance as seen by the constant behavior during charging in the CC period. In the CV period, the anode electrolyte resistance increases and dominates the cell resistance. The portion of the CV period in the slow charging is shorter (the CC period is longer) compared

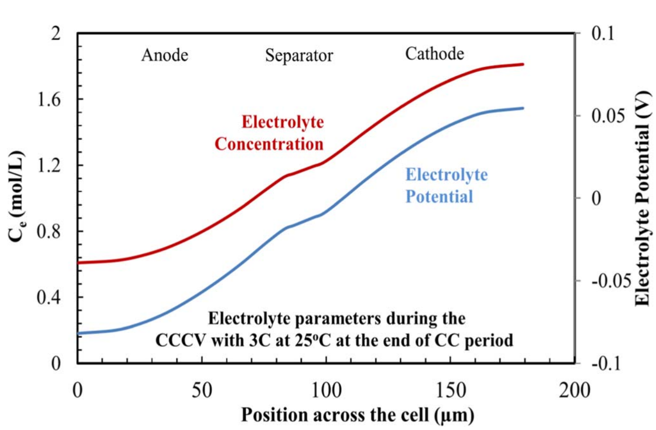

Figure 5. The electrolyte concentration and potential profile across the cell at the end of the constant current charging period for fast charging starting at 25°C with charging rate of 3C and a heat transfer coefficient of 28 W/m^2.K.

to fast charging due to the low concentration and potential polarization. Beside the charging rate, the influence of thermal condition of the cell can be studied by comparing the resistance from Figure 4(a) and 4(d). The adiabatic condition ($h = 0$) outperforms the self-heating cooling conditions in terms of cell internal resistance. In the adiabatic condition, the heat generated inside the battery during charging increases the temperature of the cell, which increases the diffusivity and conductivity. The increase of temperature also results in a decrease in the charge transfer kinetic and solid particle resistances. It is worthy to noticing that, while the performance of the cell is improved, the raising temperature may exceed the safety level (~55°C) of the cell and could produce an inhomogeneous temperature distribution inside the cell. The self-heating exhibits an option to decrease the charging time and improve the efficiency as long as the temperature is controlled below the safety threshold. On the opposite, the isothermal condition has no safety issue but the charging time is longer than the self-heating condition. In general, the cell resistance which affected by thermal operation condition (i.e. temperature and cooling conditions) is the key to improve cell charge performance.

As shown in Figure 2(e) for subzero temperatures, a lower range of C-rates are employed due to the high internal resistances that cause shutdown when charging at C-rates larger than 1 C. Comparing with the moderate temperature (25°C), a steeper reduction in the columbic efficiency is observed at low temperatures (Figure 2(f)). In the discharge process at subzero temperatures, the capacity loss is caused by the high electrolyte resistance at the isothermal condition and the

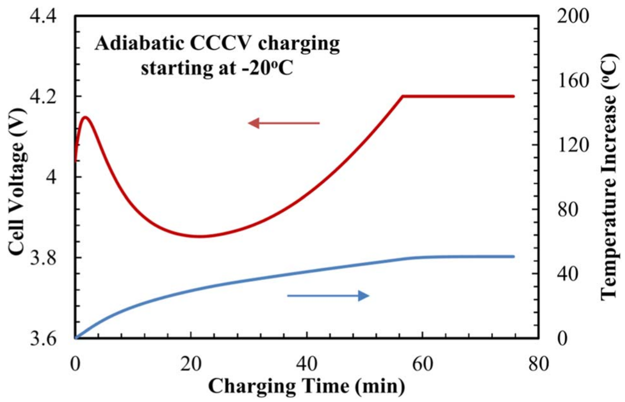

Figure 6. The charging performance curve and the temperature increase of CCCV adiabatic thermal condition (heat transfer coefficient of 0 $W/m^2.K$) charging starting from −20°C with charging rate of 1C.

solid state diffusion resistance at the adiabatic conditions[16] However, as shown in Figure 2(f), a smother decrease in the cell efficiency at adiabatic and convective heat transfer conditions is observed. The enhanced behavior is mainly due to the rise of temperature as shown in Figure 3(b), which increases the ionic conductivity and diffusivity. Similar to the influence to the coulombic efficiency, a high difference in the charging time is observed at different thermal condition as shown in Figure 2(e). The adiabatic condition manifests a considerable charging time reduction compared to other cooling conditions. The variation in the charging time and performance of the cell at different thermal condition are examined based on the cell internal resistances at 1C charging rate. As shown in Figure 4(e) and 4(f), the anode interfacial kinetics resistance dominates the cell resistances. The predicted results obtained by the present model, that interfacial kinetics resistance is the rate limiting factor of low temperature charging, agrees with experimental observations.[11] The effect of the electrolyte resistance on the short CC period is understood by the sharp concentration gradient in the electrolyte. The combined effect of slow kinetics at low temperature and the high concentration polarization brings the cell to the upper voltage limit (4.2 V) rapidly, that diminishes the duration of CC period and extends the CV period significantly. Figure 4(f) shows the decrease of cell resistances in the case of adiabatic charging. The interfacial kinetics resistance decreases with charging due to the rise in cell temperature. The temperature rise successfully suppresses the interfacial kinetics resistance of the battery, which is the dominated cell resistance in the isothermal condition. Beside the decrease of the interfacial kinetics resistance, electrolyte and solid diffusion resistances also decrease at adiabatic condition. A uniform active material utilization is predicted based on the solid phase diffusion resistance curve, which reveals two fluctuations. These fluctuations propose that the electrode is near its equilibrium state and minimum resistances. The electrolyte resistance shows two different trends during the charging process. Initially, a sharp increase in the electrolyte resistance at the anode and cathode takes place. Afterwards, the resistance reaches a peak in which it starts to diminish until the end of charge process. The difference between cell resistances is responsible for the decay in the voltage in the adiabatic conditions as shown in Figure 6. At the initial stage during adiabatic charge ($t < 20$ min), due to the high interfacial kinetics resistance, the flux of ion in the electrolyte is higher than the electrochemical reaction rate (interfacial kinetics). The difference caused the accumulation of Li^+ in the electrolyte on the anode side. The accumulation of Li ion decreases the voltage as we can see from Figure 6. At the second stage ($t > 20$ min), since the increase of temperature decrease the interfacial kinetics resistance and the Li^+ stop accumulating, the voltage start to increase until the end the CC period. In the CV period, the electrolyte resistance starts to dominate since the interfacial kinetics resistance drop due to the increase of temperature as shown in Figure 4(f).

Charging protocols.—Moderate temperature charging protocols.— The charging protocols have investigated in this subsection for fast charging applications. Most of the previous studies have focused on finding protocols at moderate charging temperatures. However, only some studies have applied or proposed the same concept for subzero and high temperatures. Moreover, there is lacking in understanding the discrepancy of the physical behavior of the battery when different charging protocol is exploited. Hence, it is worthy to explore the battery's behavior and performance under different charging methodologies. It is important to note that, in this work our focus is on studying the effects of different charging methods on the physical behavior and charging performance. The detailed discussions of the parameters of charging protocol, such as voltage limits and phase time limits, are not included in this work and left as a future exercise. The physical comparison presented in this work includes: (a) the difference between constant current (CC), constant voltage (CV), constant current constant voltage (CCCV), and boostcharging protocols; (b) the influence of the charging current in the CC, CV, CCCV, and boostcharging protocols on the charging performance; and (c) the advantage of using pulse current. At the end, the influence of a combined protocol is investigated in order to improve charging performance under subzero and high temperature operations.

Among the studied charging protocols in the literature *Constant Current (CC) protocol* is considered the simplest to execute from a practical standpoint.[33] Figure 7(a) displays the charging time and charging efficiency with different charging rates (C-rates) of CC charging. Clearly, the increase of the charging rates accomplishes a reduction of the charging time of the cell. However, the reduction of charge duration is achieved at the expense of the charge efficiency. Typically, the CC protocol is followed by a CV charging where lower charging rates is imposed to overcome the loss of the charge efficiency which extend the charging time due to the high resistances as discussed previously. Figure 7(b) shows the local utilization of the active material across the cell which is introduced with 3C-rate charging at 25°C. The figure illustrates the material utilization during charging process. It is observed that the anode is mainly utilized in the region next to the separator. The high utilization near the separator is ascribed to the high electrolyte resistance that prevail the cell resistance during charging. Also, at the surface of active material, the lithium saturates by reaching the maximum concentration value that causes the cutoff of the charging process in a short time after the start of charging. Accordingly, a key factor that controls the charging performance and achieves fast charging with a slight impact on the charge efficiency is a crucial need for charging protocols to compete with the CC protocol. A successful protocol would bring the lithium concentration to a high level without reaching the saturation limit.[5,6,34]

The pulse charging protocol with varying currents have been adopted in this study to compare with the conventional CCCV protocol in terms of charging times, efficiency and temperature increase. The varying current pulse charging mode have been proposed by Purushothaman and Landau.[6] It consists of constant width charging and rest pulses and the applied current density is decreasing with charging progress as shown in Figure 8(c). The simulation results of the pulse charging protocol with an average of 2C rate charging and convective cooling conditions for with and without CV period scenarios are shown in Figure 8(a) and 8(b), respectively. The temperature rise is about 10°C in the pulse charging protocol due to the high charging rates at the foremost stages of charging. The effectiveness of the pulse charging mode is based on the concept of increasing the surface concentration of the lithium at initial stages near the saturation value then maintaining the concentration values close to the saturation. This is achieved by exciting the cell via the high charging rates at the beginning that brings the cell voltages near the maximum voltage, then stabilizing the cell, which is manifested by the decreasing charging rates, which sustain the cell voltage level. The rest period in the pulse can prevent the saturation of lithium ion on the surface of active material, which can maintain a high CC charging rate. Figure 9 illustrates the interface State of Charge (iSOC) at the anode/separator interface during the pulse charging. From Figure 9, the iSOC increases rapidly

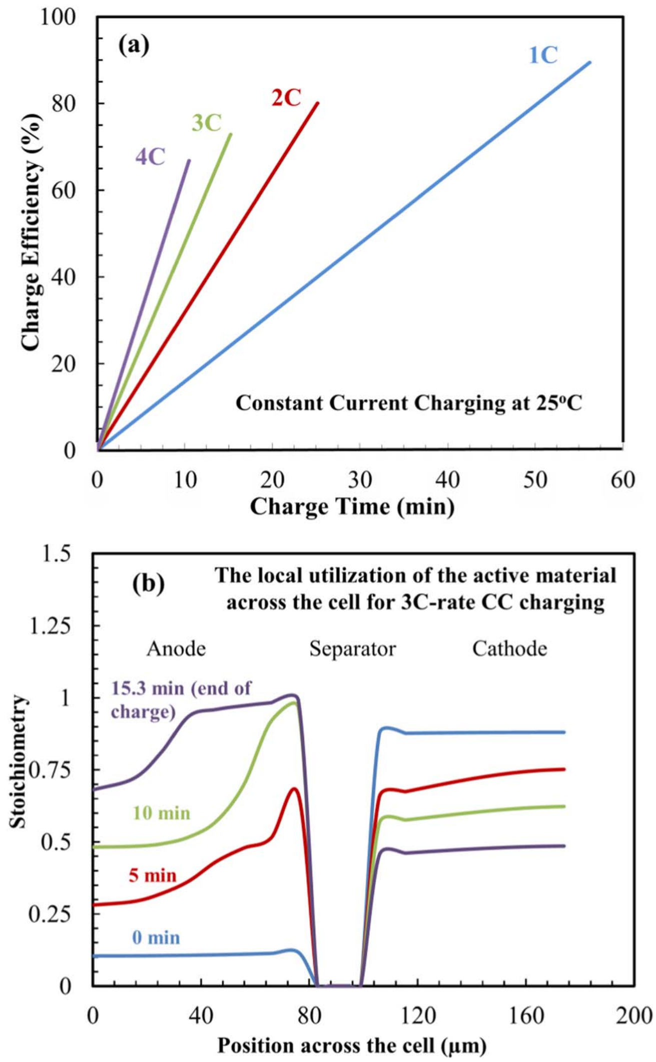

Figure 7. (a) The charging time and the charging efficiency (state of charge) during charging with constant current protocol with different applied charging currents starting at an operating temperature of 25°C and a heat transfer coefficient of 28 W/m^2.K. (b) The resultant local utilization of the active material at different times during charging across the cell for constant current charging at 3C-rate starting at 25°C.

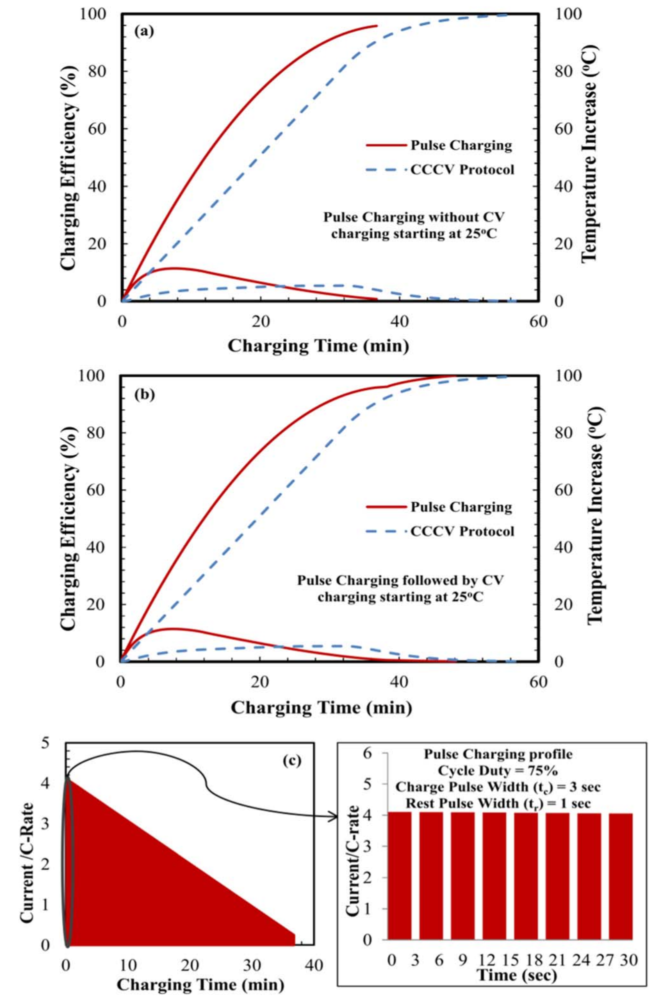

Figure 8. (a) Comparison between the decay pulse charging protocol with an average applied current of 2C and CCCV with 2C protocol in terms of the charging time, capacity return and temperature increase. The operating initial temperature is 25°C and the heat transfer coefficient value is 28 W/m^2.K. (b) The decay pulse charging protocol is followed with a constant voltage of 4.2 V to bring the charge capacity return. (c) The configuration of the decay pulse charging protocol used in the simulations. The duty ratio is 75%, the charge pulse width is 3 sec and the rest pulse width is 1 sec.

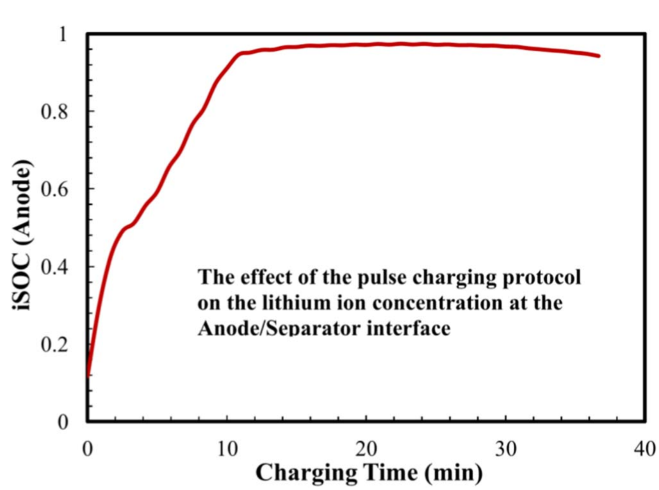

Figure 9. The normalized lithium ion concentration at the anode/separator interface (iSOC) during charging the battery with the decay pulse charging protocol with an average applied current of 2C with an operating temperature is 25°C and a heat transfer coefficient of 28 W/m^2.K.

since the average C-rate is high at the beginning. At this stage, the anode is mostly utilized in the region near the separator and causes non-uniform active material utilization inside the anode. In later stages of pulse charging, the reduction of the charging current and the rest pulses prevent the lithium concentration from exceeding the saturation value, which maintains the lithiation process in the anode. Although pulse charging decreases the charging time, the charge efficiency is lower (∼95%) compared to the CCCV charging, which can reach 100% charging efficiency with a longer time as shown in Figure 8(a). To increase the charge efficiency of the pulse charge, the charging efficiency can be increased by adding CV charging after the pulse charging. The charging efficiency of pulse-CV charging protocol is as shown in Figure 8(b). The CV period leads to a higher degree of lithium intercalation to the anode from the electrolyte. From the results, a high effectiveness of the varying pulse charging is attained with respect to fast charging and charging efficiency suggesting a successful charging protocol.

Along with pulse charging protocol, a variety of innovative charging protocols have been proposed to accomplish charging time reduction along with high charging efficiency. The basic principle of these protocols is to bring the cell voltages to the upper maximum voltage

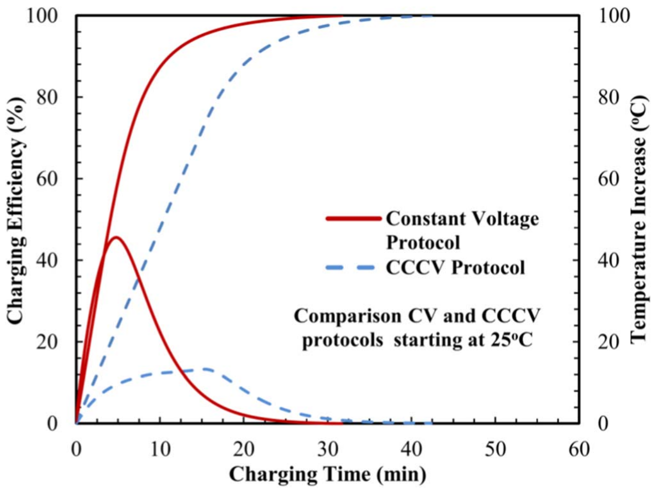

Figure 10. Comparison between constant voltage protocol with maximum applied voltage of 4.2 V and the conventional CCCV charging protocol with 3C applied current in terms of the charging time, capacity return and temperature increase. The operating initial temperature is 25°C and the heat transfer coefficient value is 28 W/m^2.K.

by applying high charging rates at the beginning of charging. The most attractive protocol respecting the charging time and efficiency is the *Constant Voltage (CV) charging protocol*. A comparison between the CV with conventional CCCV protocols simulation results of charging time, efficiency and temperature increase inside the battery are shown in Figure 10. It can be seen that the CV charging exceeds the conventional charging and achieves a high charging time reduction. At the beginning of the CV charging, the applied charging rates are exceptionally high due to the low internal resistances inside the cell. With the progression of the charging process, the increase of the internal resistances causes the charging current to decrease rapidly in order to maintain a constant voltage operation. Despite the significant time reduction accomplished by the CV protocol, it is infrequently used since the need of high power supplies to provide the high charging current at the onset of charging. Moreover, the high temperature inside the battery that resulted from the high current may have safety concern.

For that reason, an innovative *Boostcharging protocol* has been proposed by Notton et al.[5] In this protocol, a boostcharge period is applied before the conventional charging, that is a high constant current imposed to bring the cell to its maximum. Figure 11 illustrates the simulation results of the boostcharging protocol with I_{max} of 5C and V_{max} of 4.2 V (Figure 11(a)) and 4.3 V (Figure 11(b)) with an arbitrary chosen boostcharge period of 7 min followed by conventional charging with 1C and 1/20C cutoff current. In the boostcharging protocol, approximately 85% of the capacity return occurs in the first 10 minutes of charging followed by charge completion in less than 40 minutes. The felicitous performance of boostcharging is manifested in improved internal cell behavior. Figure 11(c) and 11(d) demonstrates that during the first constant current stage ($t < 7.6$ min), a high concentration and potential gradient occurs inside the cell. The high electrolyte concentration gradient causes the deviation of the ionic conductivity and diffusivity from the ideal values. The constant voltage charging ($t > 7.6$ min) during the boostcharging is a key factor in the superior battery performance during this protocol. The constant voltage reduces the electrolyte concentration polarization and voltage loss by decreasing the charging current. Furthermore, due to the lower ohmic drop, a better active material utilization at the anode side is predicted, which consumes the electrode equally rather than the

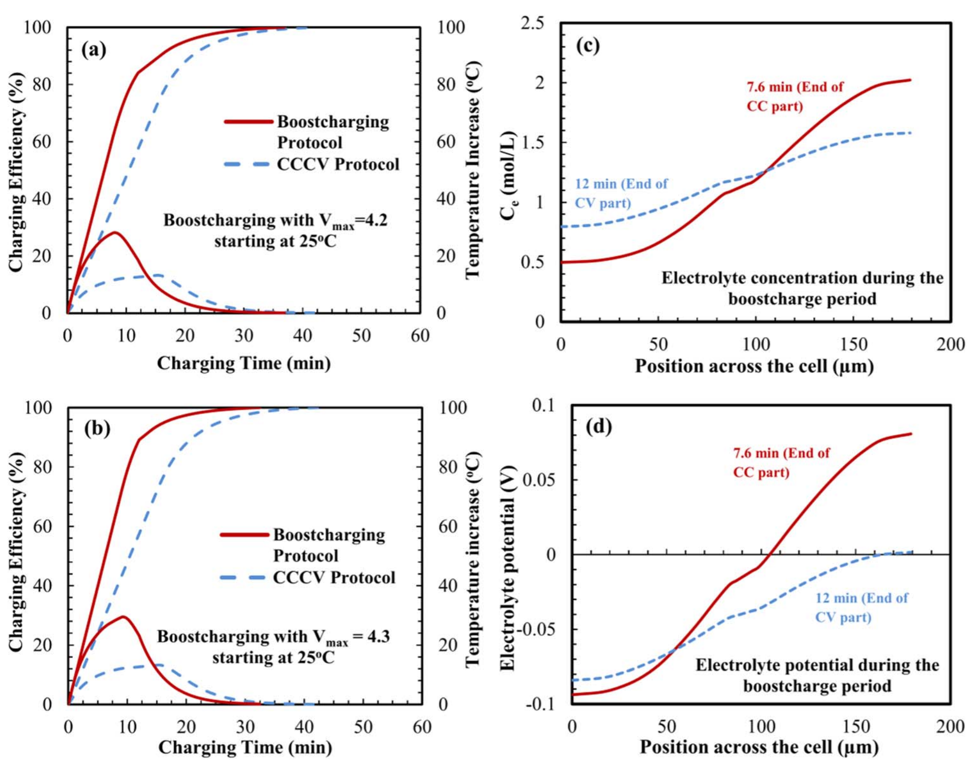

Figure 11. (a) Comparison between the boostcharging protocol in which the boostcharge period consist of constant current of 5C for 7.6 min and constant voltage of 4.2 V for 4.4 min followed by the conventional CCCV with 1C and 4.2 V protocol and the conventional CCCV with 3C protocol in terms of the charging time, capacity return and temperature increase. The operating initial temperature is 25°C and the heat transfer coefficient value is 28 W/m^2.K. (b) The boostcharge period consist of constant current of 5C for 7.6 min and constant voltage of 4.3 V for 4.4 min followed by the conventional CCCV with 1C and 4.2 V protocol. (c) The electrolyte concentration profile across the cell at the end of the constant current (5C) and constant voltage (4.2 V) in the boostcharge period. (d) The electrolyte potential profile across the cell at the end of the constant current (5C) and constant voltage (4.2 V) in the boostcharge period.

accumulation at the anode/separator interface as shown in Figure 7(b). In the case of $V_{max} = 4.3$, the charging time reduction is better than the boostcharging protocol with $V_{max} = 4.2$ due to the higher charging rates permitted in the boostcharging period. The electrochemical thermal model also provides a confident response on the thermal behavior and temperature distribution inside the battery. The temperature rise during fast charging is a critical characteristic and could not be abandoned in contrast to the isothermal model, which provides misleading results.

Low temperature charging protocols.—The cell resistance during low temperature operation is significantly higher than the room temperature operation. The applied fast charging protocols at room temperature might not be appropriate for low temperature charging due to the difference in the internal nature of the cell. Similarly, the charging currents rates employed in the case of low temperature operation should be smaller.[12]

In this study, the charging rates utilized are ranged from 0.2C to 1C for the simulations of constant current charging as shown in Figure 12(a). Clearly, as the charging rates increase, the charging time decrease, however, a substantial drop in the charging efficiency. The poor performance of low temperature CC charging is attributed to the sluggish kinetics and the concentration gradients in the electrolyte. Figure 12(b) and 12(c) show the reaction current and electrolyte concentration at different times during 1C charging at −20°C. The low charging efficiency at 1C-rate can be explained by the low reaction rates and the small ionic current that travels from the cathode to the anode because of the high ionic transport resistance. Due to the low flux, a considerable concentration gradient arises at early time of charging in the electrolyte and inducing the cell shutdown. The CC protocol is not adequate for low temperature charging, especially at high charging rates, which should be followed by the CV charging to increase the charging efficiency that results in an extended charging times as discussed in the previous section.

Limited charging protocols have been introduced for low temperature applications. We propose a subzero temperature charging protocol that consist of a decay pulse current followed by CCCV charging. Figure 13(a) shows the charging performance curve of the pulse-CCCV protocol. The charging efficiency and temperature increase with adiabatic thermal boundary condition results are shown in Figure 13(b). In this protocol, a high pulse current is applied intermitted with rest pulses followed by the conventional CCCV protocol. The proposed mode is based on the concept of battery thermal excitement; via charge pulses followed by the conventional charging protocol. That is to say, benefiting from the temperature increase during the pulse, the electrochemical properties of the cell are improved which enabling a better low operating temperature charging performance. In principle, the advantageous consequence of the pulse charging period is that it thermally excites the cell and maintains a high charging rate. During the pulse period, the cell temperature increases which enhances the charge transfer kinetic and lithium diffusion in the electrodes. The rest pulses prevent the development of a high polarization inside the cell and large anode overpotential, which keeps the cell voltage below the upper voltage limit. It is evident that, the heat generation warm up the battery and efficiently reduces the cell resistance. The heating of the cell during low temperature operation has demonstrated to contribute with faster and enhanced charging efficiencies, as illustrated by the adiabatic and proposed pulse-CCCV charging protocol.

Influencing factors in fast charging operation.— Electrode design parameters.—Several studies approached the design parameters of the cell to improve its performance on account of the significant effect on the electrochemical characteristics. However, the majority of these studies have focused on the discharge performance.[35] Limited studies have addressed the performance during fast charging at different operating temperatures and cooling condition. The parametric study is performed for the subzero (−20°C), moderate (25°C) and high temperatures (45°C) to identify the charging time and charge capacity for each design parameter. The CCCV charging protocol with

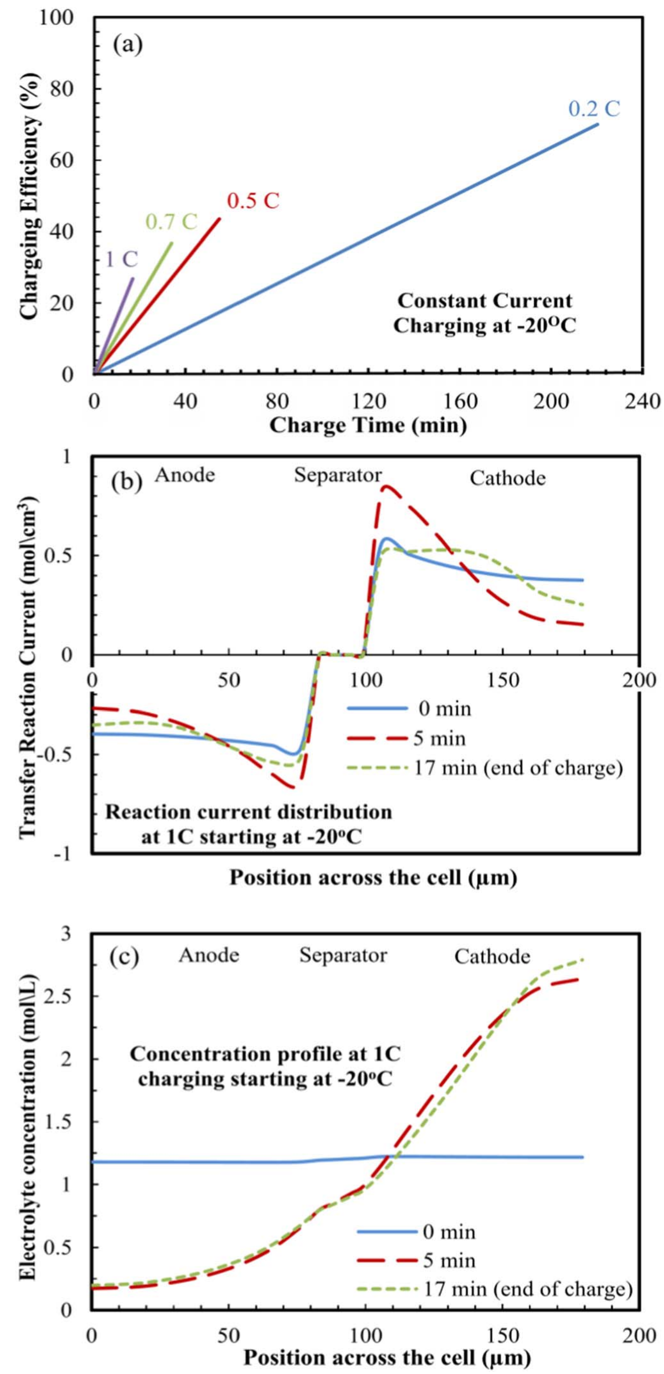

Figure 12. (a) The charging time and the charging efficiency (state of charge) during charging with constant current protocol with different applied charging currents starting at an operating temperature of −20°C and a heat transfer coefficient of 28 W/m^2.K. (b) The local transfer reaction current profile at different times during charging across the cell for constant current charging at 1C-rate starting at −20°C. (c) The electrolyte concentration at different times during charging across the cell for constant current charging at 1C-rate starting at −20°C.

maximum voltage of 4.2 V and 0.05C cutoff current are employed to conduct the parametric analysis with charging currents of 3C for room temperature and 1C for subzero temperature. The cell design parameters considered in the analysis are the electrode thickness, electrode porosity, and particle size. The separator thickness and porosity are held constant at the base case parameters as shown in Table I. The influence of the anode thickness is studied by changing its value from

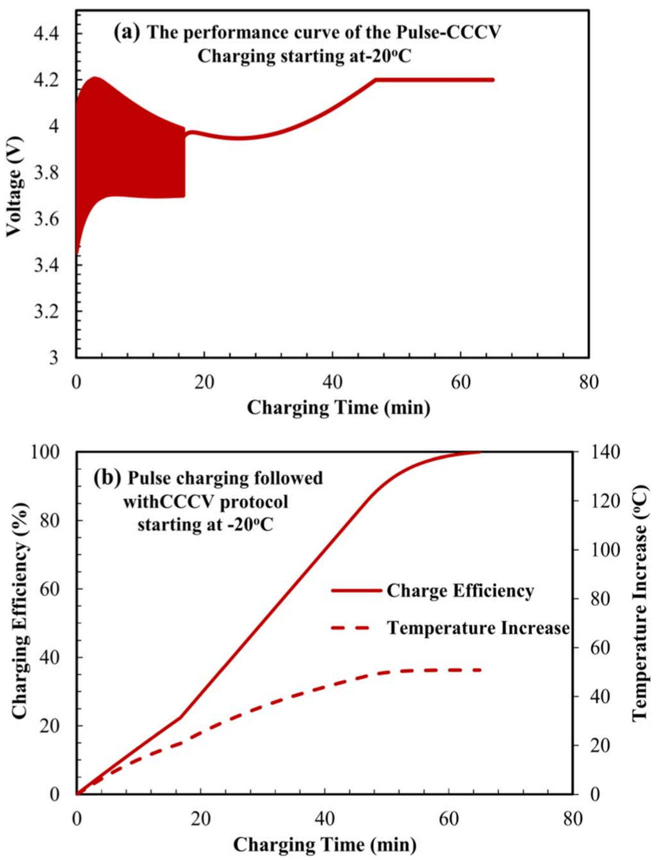

Figure 13. (a) The charging performance curve and the temperature increase of the pulse-CCCV protocol with charging rate of 1C starting at an operating temperature of $-20°C$ and a heat transfer coefficient of 0 $W/m^2 \cdot K$ (adiabatic thermal condition). (b) The charging time and efficiency (state of charge) during charging with the pulse-CCCV protocol.

56 μm to 101 μm. For the anode porosity study, the porosity of range 0.2 to 0.5 is used to capture its influence. The result of the parametric studies and parameters influences on the fast charging performance are illustrated in Figure 14. It demonstrates the tradeoff between the charge capacities and the charging time at 25°C as a result of varying the design parameters. As observed, the anode parameters effectively impact the cell performance with respect to charging time and charge capacity.

Electrode porosity.—As shown in Figure 14(a), the charge capacity of the battery is decreasing when increasing the anode porosity. The charging time, shows the same trend with the increase of the anode porosity. The decline of the charge capacity with porosity is because of the decrease of the electrode active material. One should notice that, the charge time increase at low porosity anode. This is because the effective ionic diffusivity and conductivity in the electrolyte in the low porosity electrode is low according to the Bruggeman relation $X_{eff} = X\varepsilon^{1.5}$. The low effective properties increase the potential and concentration polarization in the cell, which decrease the CC period (increase the CV period) and total charging time. The predicted charging performance parametric studies for anode porosity at $-20°C$ and 45 °C show generally analogous trend as shown in Figure 15(a) and 16(a). The increase of charging time becomes severe at $-20°C$ since the low temperature further decrease the effective properties. On the opposite, the change of charging time at 45 °C is less than 25°C. Apparently, at high temperature, availing from the room temperatures in the cell, the effective electrolyte parameters, and the charge transfer kinetics in addition to the solid state diffusion all show improvement in their cell performances.

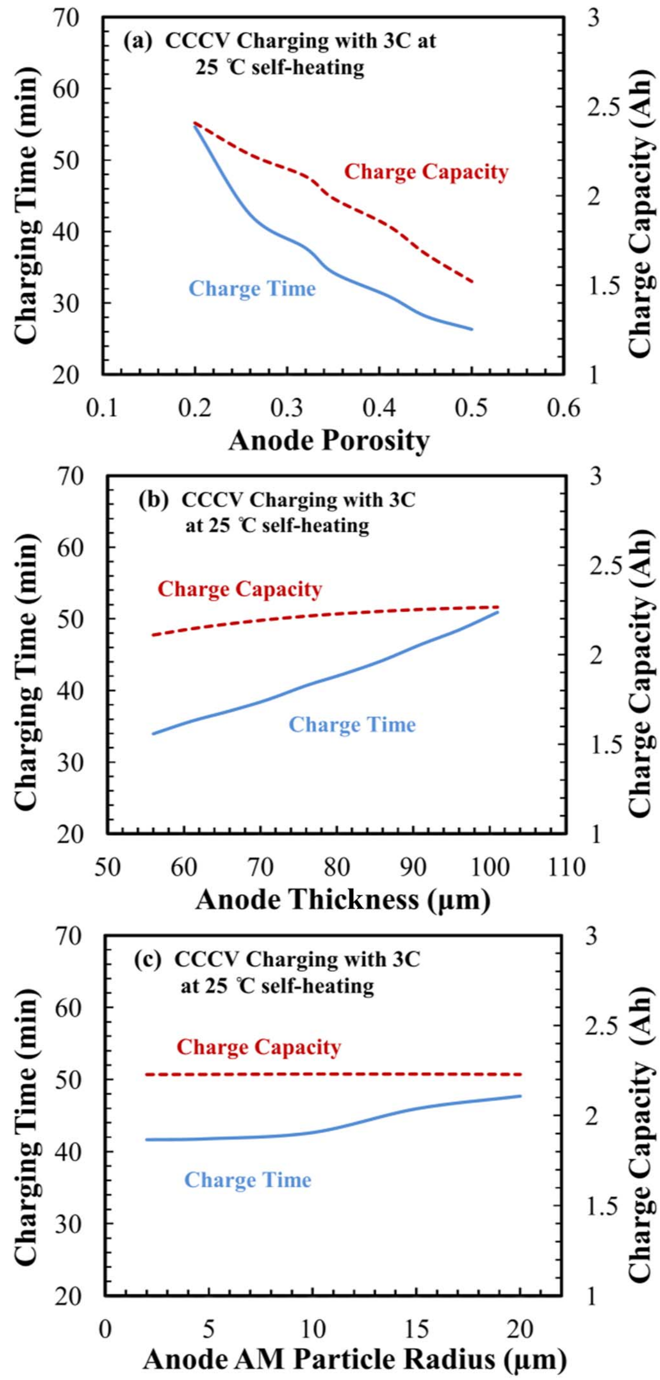

Figure 14. The effect of the microstructural electrode design parameters on the charging performance (charging time and capacity) with using the conventional CCCV charging protocol with 3C. The operating initial temperature is 25°C and the heat transfer coefficient value is 28 $W/m^2.K$. (a) Anode porosity. (b) Anode thickness. (c) Anode active material particle size.

Electrode thickness.—The anode thickness on the other hand, reveals comparative relation between the charging time and the charge capacity as illustrated in Figure 14(b). The charge capacity is superior for thicker anodes concurrently longer charging time is sought. The battery charging performance maintains a similar trend for both high and low temperatures as seen in Figure 15(b) and 16(b) with different charging time values. The observable details outcomes an optimal range of design parameters that yield in the minimum charging time with a high charge capacity. The results of the parametric study are

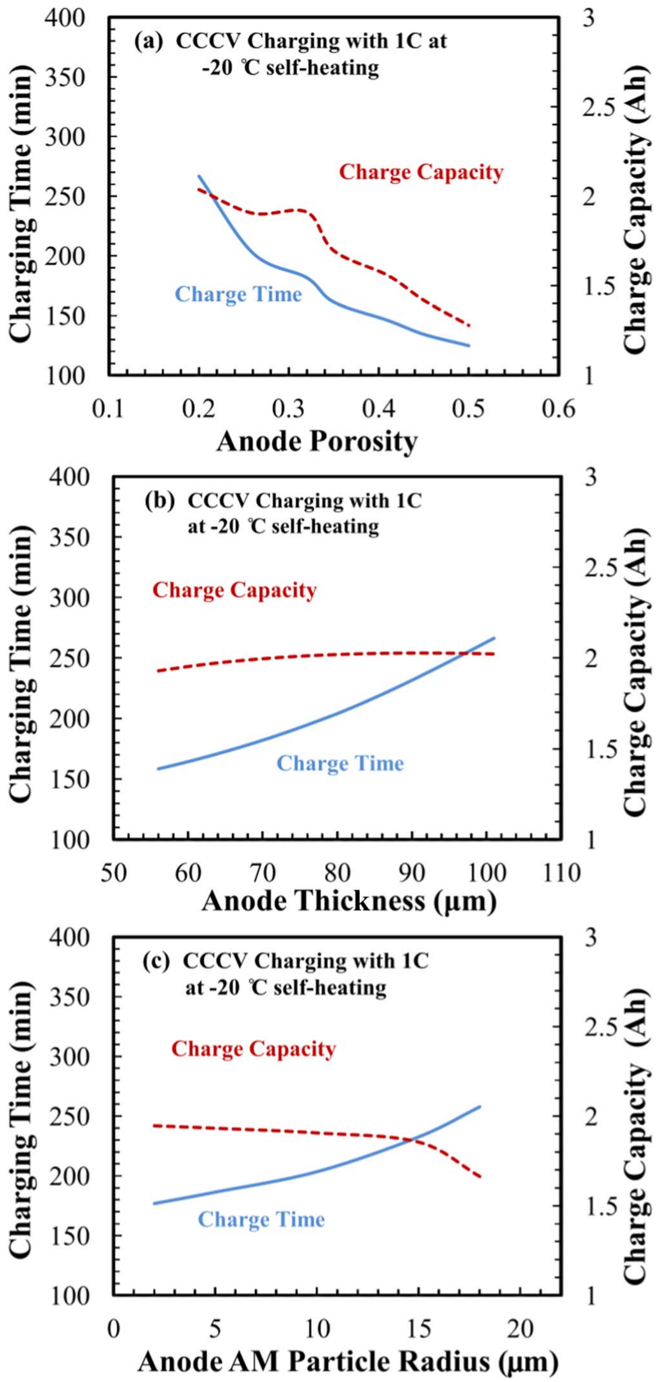

Figure 15. The effect of the microstructural electrode design parameters on the charging performance (charging time and capacity) with using the conventional CCCV charging protocol with 1C. The operating initial temperature is −20°C and the heat transfer coefficient value is 28 W/m^2.K. (a) Anode porosity. (b) Anode thickness. (c) Anode active material particle size.

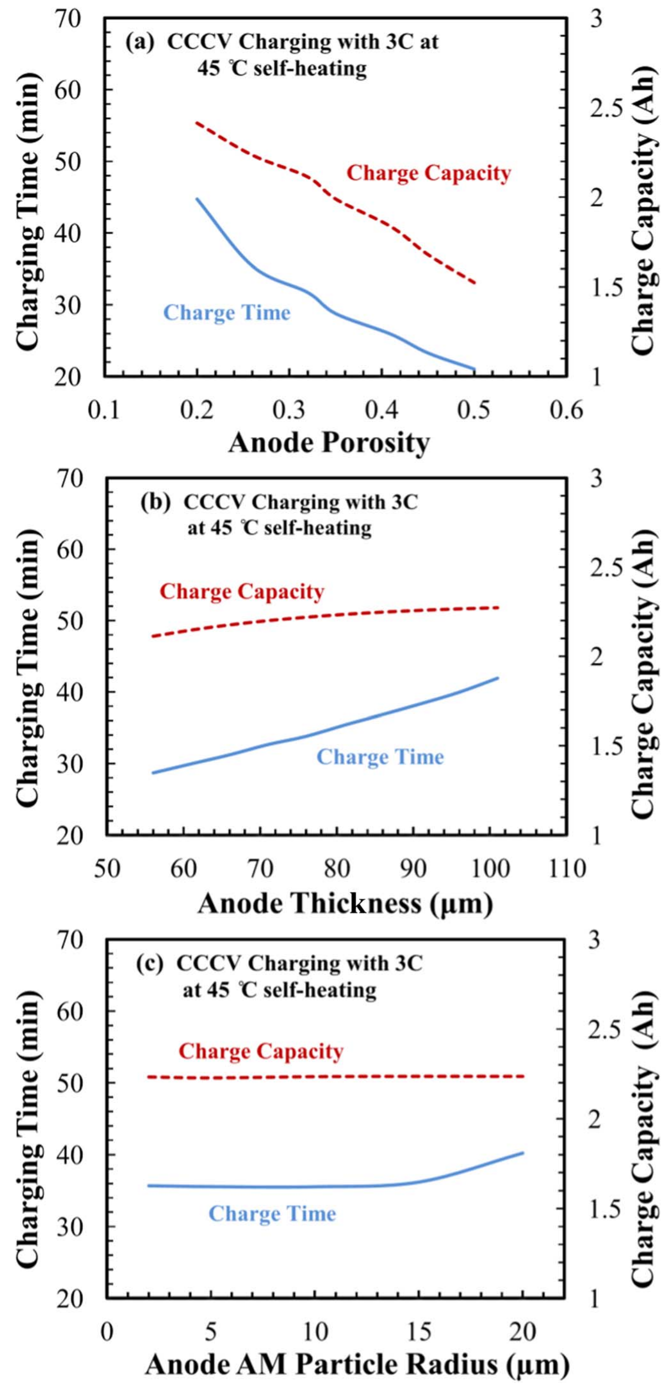

Figure 16. The effect of the microstructural electrode design parameters on the charging performance (charging time and capacity) with using the conventional CCCV charging protocol with 3C. The operating initial temperature is 45°C and the heat transfer coefficient value is 28 W/m^2.K. (a) Anode porosity. (b) Anode thickness. (c) Anode active material particle size.

compatible with theoretical simulation of charge performance predicted by an electrochemical model.

Understanding how the thickness and porosity affect the charge performance is recognized by studying the limiting mechanisms in the electrodes. This is manifested in the reaction current density across the cell that relates the electrochemical reaction and the pore wall flux as shown in Figure 17. The reaction current density at the end of the CC charging period for different thicknesses and porosities are shown in Figure 17(a) and 17(b) respectively. A peak subsists in the reaction current that represents the maximum reaction rate and active material utilization. The transfer current density next to the separator becomes poorer with time because the active material particles in this region are almost completely charged, hence, the peak moves to the current collector to satisfy the charge balance equation. The peak shifts from the anode/separator interface to the end of the anode during the CC charging period. Therefore, in order to achieve a high active material utilization, the peak of the transfer current density should reach the anode current collector at the end of CC charge. The reaction current density peak location has a direct relation with the cell design

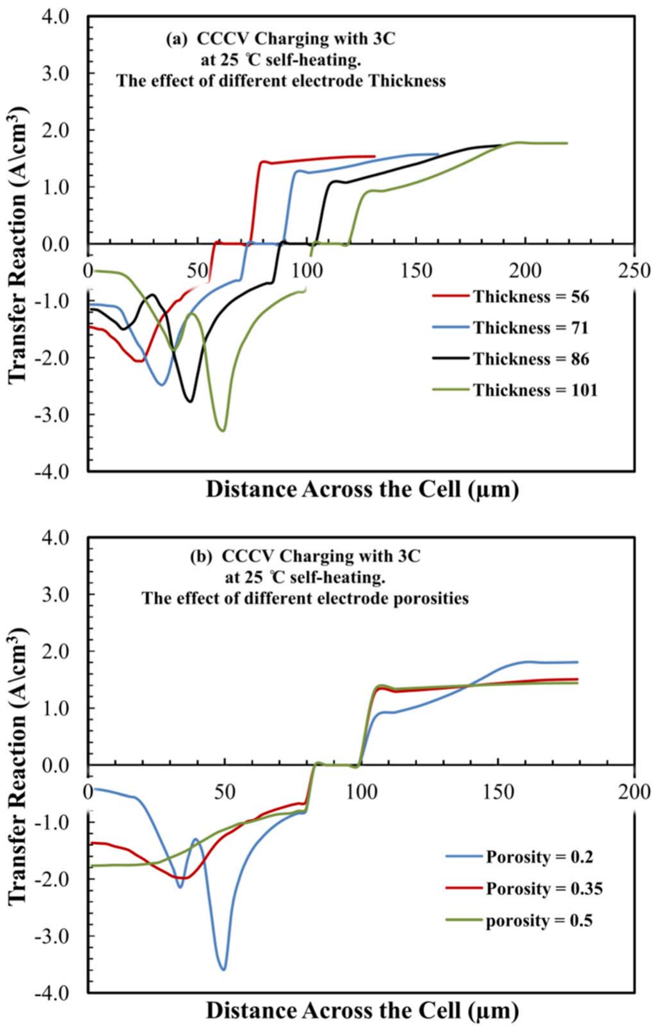

Figure 17. The transfer reaction current distribution across the cell at the end of the CC period when charging with conventional CCCV with 3C. The operating initial temperature is 25°C and the heat transfer coefficient value is 28 $W/m^2 \cdot K$. (a) The distribution at different electrode thicknesses. (b) The distribution at different electrode porosities.

parameters that influence the charging performance.[36] In the case of thin electrode thickness, the ionic current path are shorter, therefore, the reaction current peak penetrates deeper inside the electrode due to the lower polarization. Therefore, the peak in the reaction current density reaches nearly to or at the current collector with the end of the CC period. The underlying reaction current penetration that shows a uniform and fast active material utilization elucidates the improvements in battery's charging times and performance for thin electrodes. For thicker electrode thicknesses, the ionic current path are longer, which causes high electrolyte resistance and low reaction current peak penetration in the anode that prolongs the charging time of the battery. For the electrode porosity, the charging time of the battery decreases with the increase of the porosity. In the case of higher electrode porosity, the effective electrolyte conductivity and diffusivity are improved, which leads to the enhancement of reaction current peak penetration inside the anode as shown in Figure 17(b). At the microstructure level, the lithium ions transport through a convoluted path at low porosities as opposed to high porous electrodes shows a localized transport reaction near the anode/separator interface and low penetration rate. The local lithium consumption creates a high concentration build up near the anode/separator interface that shifts the charging to the constant voltage period.

*Active particle size.—*Beside the porosity and tortuosity, the effect of the anode active material particle size on the charging performance has been studied. The particle size have less impact on the battery charging time and charge capacity on high and moderates operating temperature charging as seen in Figure 14(d) and 16(d). On the other hand, as shown in Figure 15(d), the lower particle sizes provides a boost in charging performance at low temperature charging. Since the interfacial kinetics resistance is the limiting factor at low temperature charging, the particle size reduction is an efficient approach to increase the battery performance that decrease the charging time and increase the charge capacity. The particle size increases the electroactive surface area of the electrode that enhances the electrochemical kinetics at the electrode surface that reduces the interfacial kinetics resistance. The increase of the particle size could increase the anode solid phase diffusion distance that will add up with the interfacial kinetics resistance to decrease the battery's performance. In summary, it is found that the design parameters have a high impact on the charging performance of the cell. In the manufacturing process, the design variables can be fabricated, which spurs the optimization of the design parameters during the process to afford the best possible performance for fast charging applications.

In order to elucidate the trade-off between charging time and capacity, the normalized charging time (min/Ah) for different electrode design parameters is presented in Figure 18. From the results it is found that the normalized charging time increases with the decrease of electrode porosity, the increase of electrode thickness, and the increase of the active material particle size, except for the case of CCCV charging at 1C and T = −20°C. Generally speaking, the decrease of porosity, the increase of ionic transport length in the electrolyte, the increase of solid-phase diffusion length in active material, and a decrease in temperature result in a prolonged normalized charging time. The existence of an optimal design parameter, namely the porosity at T = −20°C suggests that the capacity lost is significantly high even with a high-porosity electrode.

Conclusions

Investigations of cell performance and limiting mechanisms during fast charging operation of Li-ion cell have been conducted at moderate and extreme operating temperatures. The simulations of the fast charging at different temperatures and thermal conditions were performed by an electrochemical thermal coupled model. In our study, charging time and columbic efficiency are found to be highly dependent on the charging rates, cooling conditions and temperature. At subzero temperatures, due to the charge transfer kinetics limitations, poor electrolyte transport and low solid-phase diffusivity, a high charging time and low columbic efficiency are found. Different from the subzero condition, the electrolyte resistance dominates the charging process at high and moderate operating temperatures. At the adiabatic conditions, the increase of the cell temperature suppresses the kinetics limitations and the electrolyte resistance, which are the limiting factors for low temperatures, and leading to a low charging time and high coulombic efficiency compared to isothermal condition.

For the charging protocols, pulse charging protocol achieves a fast charging by increasing the surface concentration of the lithium at initial stages. Then, near the saturation value, maintaining the concentration values close to the saturation concentration by decreasing the pulse amplitude. For boost charging protocol, the applied constant voltage at the boostcharging period reduces the charging time. The following constant current stage decreases the high concentration gradient, which formed in the constant voltage stage, to increase the coulombic efficiency. For low temperature charging protocols, by taking the advantage from pulse charging and CCCV charging protocol, a successful reduction of time is achieved by applying pulses-CCCV charging protocol. Additionally, the electrode design parameters show that optimization of the cell design parameters improve the charging process performance by affecting the nature of the voltage losses during charging duration in the cell. The anode thickness and porosity could be effectively optimized to rapid charging, combined with high

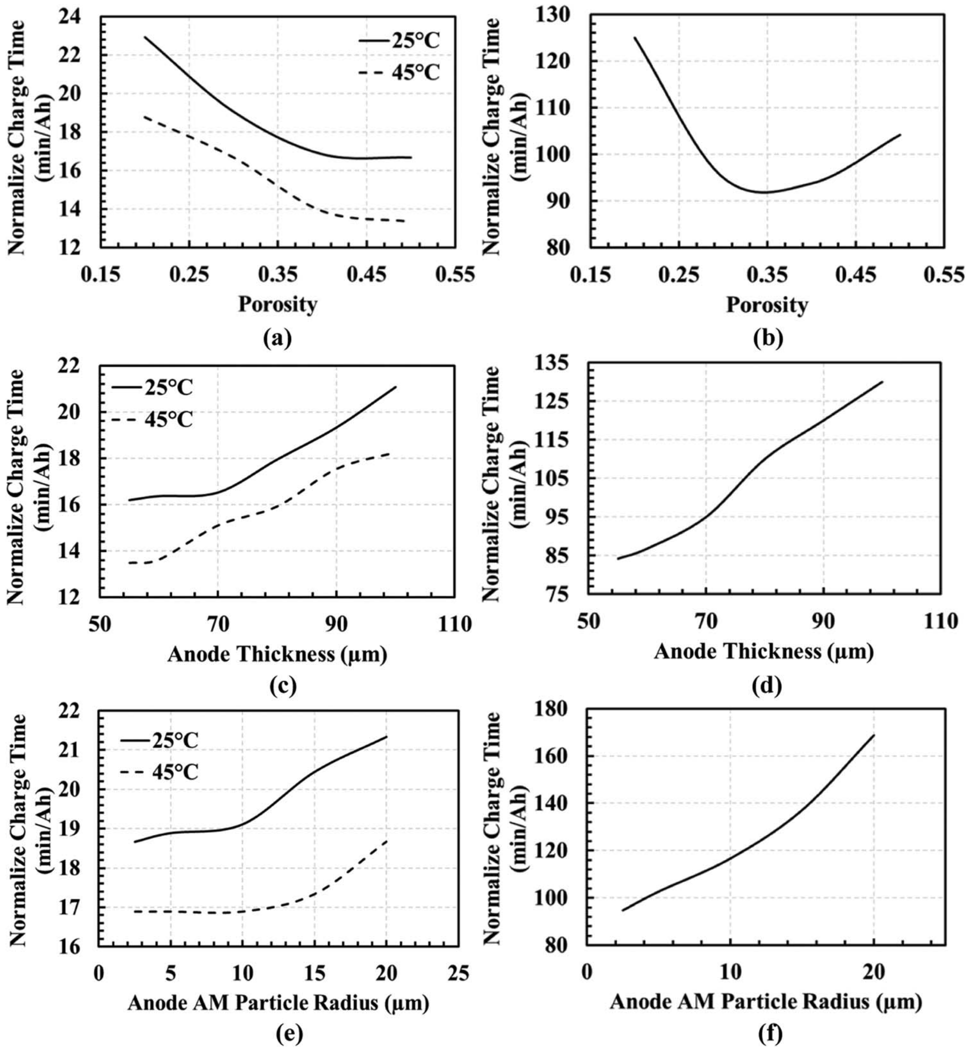

Figure 18. The effect of the anode porosity, thickness, and AM particle radius on the charging performance with using the conventional CCCV charging protocol. The heat transfer coefficient value is 28 W/m^2.K. (a)(c)(e) CCCV charging with 3C at 25°C and 45°C. (b)(d)(f) CCCV charging with 1C at −20°C.

charge capacity that can be seen by the local transfer current penetration in the anode during charging.

Acknowledgment

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List of Symbols

A_{cell}	surface area of the cell (m^2)
a_s	specific surface area (1/m)
c_e	electrolyte concentration (mol/l)
c_s	solid phase concentration (mol/l)
C_p	cell heat capacity (J/kg K)
D_e	lithium ion diffusivity in the electrolyte (m^2/s)
D_s	solid phase diffusivity (m^2/s)
$E_{a,i}$	activation energy (J/mol)
F	faraday's constant (J/mol)
h_{conv}	heat transfer coefficient (W/m^2.K)
$j_{0,i}$	exchange current density (A/m^2)
j_{loc}	local current density (A/m^2)
m_{cell}	cell mass (kg)
P	arbitrary parameter
$\dot{Q}_{act}$	active polarization heat generation (J/m^3)
$\dot{Q}_{ohm}$	ohmic heat generation (J/m^3)
$\dot{Q}_{rxn}$	reaction heat generation (J/m^3)
R	gas constant (J/mol K)
R_f	film resistance (Ω/m^2)
$R_{Particle}$	active material particle radius (m)
T	temperature (K)
t^+	transference number
T_{amb}	ambient temperature (K)
U_n	open circuit potential of negative electrode (V)
U_p	open circuit potential of positive electrode (V)

Greek

α_a	transfer coefficient for anodic current
α_c	transfer coefficient for cathodic current
ε_e	volume fraction of the electrolyte
ε_s	solid phase active material volume fraction
ϕ_i	electrical potential (V)
γ	Bruggeman coefficient
η_i	overpotential (V)

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